Document Control

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Obstetric Haemorrhage Guideline	
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Superseded Documents
Previous versions
Antepartum Haemorrhage guideline
Postpartum Haemorrhage guideline

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- Infection Control
- Medicines Management
- Anaesthetists
- Obstetricians

• Midwives • Haematologist Lead • Women's and children's Directorate Management • Womens and Childrens Pharmacist	
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1. Purpose

1.1. The purpose of this document is to detail the processes required in the event of an obstetric haemorrhage. This document will provide guidance on both Antepartum Haemorrhage (APH) and Postpartum Haemorrhage (PPH).

1.2. The policy applies to all clinicians working within the maternity services.

1.3. To discuss the specific management of conditions such as placental abruption and uterine rupture is outside the remit of this guideline.

1.4. Implementation of this policy will ensure that:

- Clinicians managing Obstetric haemorrhage have access to clear guidance which is in-keeping with national guidance and recommendations.
- Clinicians are able to identify those women at increased risk of obstetric haemorrhage and act accordingly, subsequently aiming to avoid obstetric haemorrhage wherever possible.
- Clinicians act appropriately in the event of an Obstetric Haemorrhage to reduce morbidity and mortality, and improve outcomes for women and their babies.

2. Introduction and Definitions

Obstetric haemorrhage remains one of the major causes of maternal death in both developed and developing countries and is the cause of up to 50% of the estimated 500 000 maternal deaths that occur globally each year. Though deaths from obstetric haemorrhage in the UK are uncommon, haemorrhage remains a significant cause of maternal death. Haemorrhage appears as the major cause of severe maternal morbidity in almost all 'near miss' audits worldwide. Obstetric Haemorrhage refers to both antepartum and postpartum bleeding. (RCOG & MBRRACE-UK)

2.1. Antepartum Haemorrhage

APH is defined as bleeding from or in to the genital tract, occurring from 24+0 weeks of pregnancy and prior to the birth of the baby. APH is associated with increased maternal and perinatal morbidity and mortality. It is recommended that women be advised to report all vaginal bleeding to their ante-natal care provider.

2.2. Postpartum Haemorrhage (Primary and Secondary)

Primary PPH is defined as loss of 500mls or more of blood from the genital tract within 24 hours of the birth of a baby (RCOG, 2016)
Secondary PPH is defined as any abnormal or excessive bleeding from the genital tract occurring between 24hrs and 12 weeks postnatally (RCOG, 2016).

3. Responsibilities and Roles

It is widely recognised that a multi-disciplinary approach to the management of obstetric haemorrhage can help to improve the outcomes for women and their babies. Management and treatment is initiated by the first clinician on the scene, usually the midwife providing care.

3.1. The role of the Obstetrician:

- The obstetrician has a responsibility to counsel women at significant risk of haemorrhage in the antenatal period.
- The obstetrician should be made aware of women that are known to be at increased risk of obstetric haemorrhage and should document a clear management plan in the woman's records.
- Diagnosing and managing the obstetric emergency; and communicating clearly with other members of the multi-disciplinary team.

3.2. The role of the Midwife:

- It is the midwife's responsibility to practise at all times under guidance from the Nursing and Midwifery Council (NMC). The NMC advises that all midwives must practise effectively: to deliver or advise on treatment without delay.
- Midwives have a duty to practise on the basis of the best available evidence and best practice.
- Midwives are required to communicate effectively, keep clear and accurate records and to share skill and knowledge appropriately.
- In most cases, the midwife providing care is the first clinician on the scene in an obstetric haemorrhage, therefore they must be clear in their role of detecting the emergency and summoning and initiating the appropriate support and treatment.

3.3. The role of the Anaesthetist:

- In obstetric practice, the anaesthetist is a valued skilled clinician and advisor to the Obstetrician and Midwife in managing major haemorrhage.
- More specifically, the anaesthetist plays a crucial role in maintaining haemodynamic stability and, if necessary, determining the most appropriate method of anaesthesia.

3.4. The role of the Maternity Care Assistant:

- The role of the MCA is to assist the midwives and obstetricians in collecting emergency equipment and preparing maternity theatre.
- MCA plays an important role in scribing events and recording timings in emergency situations

3.5. The role of the Haematologist and Transfusion Service:

- To manage the blood product replacements that may be required in major haemorrhage.
- To provide advice on the type of blood product to utilise and the availability of blood products in the trust.
- Communication between clinicians in the maternity unit and the haematologist is essential in such situations.

4. Antepartum Haemorrhage (APH)

4.1. Further Classified into:

- **Minor haemorrhage**: blood loss less than 50mls that has settled
- **Major haemorrhage**: blood loss between 50-1000mls, with no signs of clinical shock
- **Massive haemorrhage**: blood loss greater than 1000mls and/or signs of clinical shock

4.2. Risk Factors for APH:

- First trimester bleeding
- Intrauterine Haematoma seen on USS in the first trimester
- Maternal Thrombophilia (such as Factor V Leiden)
- Domestic Violence
- Smoking and drug abuse
- Placenta Praevia

4.3. Causes of APH include:

- **Placenta Praevia**: where implantation of the placenta has occurred completely or in part of lower segment of the uterus. There are different classifications of placenta praevia, depending on the extent to which the placenta covers in the internal os.
- **Placental Abruption:** premature separation of the placenta from the uterine wall.
- **Uterine Rupture**
- **Vasa Praevia:** bleeding from ruptured umbilical vessels lying within the fetal membranes, succenturiate placental lobe or abnormal insertion of the cord.
- Unexplained APH: Also associated with increased risk of adverse maternal and perinatal outcome.

4.4. Management of APH

Minor APH:

Women presenting with an APH must be managed in an area where there is access to a multidisciplinary team including midwifery and obstetric staff, with immediate access to laboratory, theatre, neonatal and anaesthetic services.

Women attending with vaginal (PV) bleeding (to include vaginal spotting) should receive a full antenatal assessment. This should include comprehensive history taking of pain, extent of vaginal bleeding and history of fetal movements. It may also be useful to consider previous smear results if applicable. The woman should receive a full set of maternal observations and an assessment of fetal well-being (CTG monitoring if > 26/40)
Placenta Praevia should be excluded; if necessary a referral for placental localisation scan should be made.
The woman's Rhesus Status should be considered and the appropriate action taken if she is Rhesus Negative (the Kleihauer test should be performed in rhesus D (RhD)-negative women to quantify fetomaternal haemorrhage (FMH) in order to gauge the dose of anti-D immunoglobulin (anti-D Ig) required). Kleihauer test is not a sensitive test for diagnosing abruption.
Continuous electronic fetal monitoring (EFM) should be commenced until bleeding has settled. This is due to APH being associated with fetal hypoxia and abnormalities of the fetal heart pattern.
The woman should be kept nil by mouth until bleeding is settled in view of the increased risk of needing an emergency transfer to theatre. Consider IV access.
Analgesia should be offered and administered at the woman's request.
The SAS Obstetric Doctor or Consultant obstetric review should be requested as a matter of urgency, the SBAR sticker should be utilised and a clear handover should be given to the obstetric team.
Review from the SAS Obstetric Doctor or Consultant Obstetrician should include:

- **Abdominal palpation** - to assess for tenderness or signs of an acute abdomen. A tense or 'woody' abdomen may indicate a significant abruption
- **PV examination** – a speculum examination can be useful in detecting a lower genital tract cause for the bleeding. Any women found to have abnormality of the cervix on speculum examination should be referred for colposcopic evaluation.
- **Digital examination** - should be avoided unless placenta praevia has been excluded by USS. However, if the bleeding is associated with pain or uterine activity digital examination can provide useful information on dilatation of the cervix.
- **CTG** should be performed for women presenting with APH from 26 weeks onwards (once mother is stable) to assess fetal wellbeing and to aid decision making on timing and mode of delivery.
- **USS** should be arranged to confirm placental site, and an assessment of growth liquor and dopplers should be included. USS should be carried out to establish fetal heart pulsation if fetal viability cannot be detected using external auscultation.

- Consideration should be given to the need for **antenatal corticosteroids** if between 24+0 and 34+6/40. In women presenting with spotting, where the most likely cause is lower genital tract bleeding, where imminent delivery is unlikely, corticosteroids are unlikely to be of benefit, but could still be considered.
- An on-going plan of care should be made

Investigations should include:

- The extent and physiological effects of the APH (will depend on the amount of bleeding).
- Bloods to include a full blood count and a group and save (urea, electrolytes and liver function tests should be considered).
- USS for placental localisation and to diagnose placenta praevia. Placental abruption is a clinical diagnosis and there are no sensitive or reliable diagnostic tests available. USS has limited sensitivity in identification of retroplacental clot. However, when the USS suggests an abruption, the likelihood that there is an abruption is high.
- In Rhesus Negative women manage as per above guidance for spotting

All women with APH heavier than spotting and women with ongoing bleeding should remain in hospital, at least until the bleeding has stopped. Women presenting with spotting who are no longer bleeding and where placenta praevia has been excluded can go home after a reassuring initial clinical assessment.
Consider alerting paediatric team and the Special Care Baby Unit (SCBU) as appropriate

Major and Massive APH

The management of Major APH is as for Minor APH but with the following steps as outlined below:

If the woman is presenting with a major or massive haemorrhage, an acute appraisal of maternal wellbeing should be performed and resuscitation started immediately.
It is essential to remember that the mother is the priority in these situations and should be stabilised prior to establishing the fetal condition.
The midwife is responsible for delivery of care and instigation of initial resuscitation procedures and should stay with patient, recording times and events in records.
A full and comprehensive history is essential in managing a major APH. Take history on admission as stated above. The clinical history should determine whether there is pain associated with the haemorrhage. Placental abruption should be considered when the pain is continuous. Labour should be considered if the pain is intermittent.
Note: the following steps may occur simultaneously, ensure each team member attending has a specific role in the management of the APH. For the purposes of this guideline the steps necessary when treating an APH have been separated into 5 sub-headings.

Call for help:

⇨ Pull the emergency bell and state the problem clearly to the team that are attending

- Dial 2222 and follow the emergency card instructions by the telephone. Say "emergency obstetric team" and state location.

Monitor and Investigate:

- Vital signs: respiratory rate, BP, O_2 saturations, pulse and temperature
- Obtain IV access immediately, 2 large gauge cannulae. Obtain bloods for FBC, U&Es, LFTs, coag (INR, APTT & fibrinogen) & calcium, and X-match at least four units. Specify urgent X-Match. (Group specific blood can be issued by the blood bank within 15 minutes of sample receipt)(Trust guideline – Blood transfusion policy REF)
- Insert Foley Catheter and commence strict fluid balance chart
- Weigh pads and incos – calculate estimated blood loss
- Keep Nil by Mouth
- Arrange immediate USS
- Consider the cause of bleeding (placenta praevia, placental abruption, uterine rupture, vasa praevia, placenta accreta)

Arrest the bleeding:

- Prepare for emergency transfer to theatre
- CTG abnormal, patient stable: SAS Obstetric Doctor and/or Consultant presence on Labour Ward and decision required, proceed to category 1 LSCS.

Maternal Resuscitation:

- Head down tilt or lie flat with all four limbs raised
- Oxygen at 10-15 litres/minute. Use reservoir bag and oxygen mask to deliver 100% oxygen.
- IV (until x-matched blood available):
- Plasmalyte 2 litres
- Blood – if x-match unavailable use uncrossmatched group specific blood
- As a last resort use Group O Rh D Negative blood
- Consider CVP line or arterial line

Please see Blood Transfusion procedures for guidance on administration of blood transfusion (REF)

Assess Fetal Condition:

- Fetal heart normal, patient stable, bleeding settled: USS, discuss further management on an individualised basis.
- Fetal heart absent, patient stable: USS to confirm IUD, discuss further management on an individualised basis.
- CTG abnormal, patient stable: the SAS Obstetric Doctor and/or Consultant presence on Labour Ward and decision required, proceed to category 1 LSCS.
- Patient stable, placenta praevia excluded, CTG reassuring and in 1st stage of labour: Consider facilitating vaginal delivery.

Post-operative care management is likely to be on ICU/HDU though should be assessed on an individualised basis.

Prepare for PPH post-delivery, as the presence of APH greatly increases the risk of PPH.

5. Postpartum Haemorrhage (PPH)

5.1. Classifications of PPH:

- Minor: Blood loss between 500-1000mls without clinical shock
- Major: Blood loss greater than 1000mls or clinical shock

N.B Clinicians should be aware that the visual estimation of blood loss is inaccurate and that clinical signs and symptoms should be included in the assessment of PPH. Best practice is to weigh any blood loss at the time of delivery to endeavour to calculate the most accurate measurement of blood loss.

5.2. Risk Factors for PPH

- Multiple Pregnancy
- Previous PPH
- APH
- Pre-Eclampsia
- Fetal Macrosomia
- Induction of labour
- Anaemia
- Failure to progress in the second stage of labour
- Prolonged third stage of labour
- Retained Placenta
- Placenta Accreta
- Episiotomy/Perineal Laceration
- General Anaesthetic
- Coagulopathy

5.3. Identifying the cause of the PPH

- TONE – abnormality of uterine contraction (the most common cause of PPH)
- TISSUE – retained products of conception
- TRAUMA – genital tract trauma
- THROMBIN – abnormalities of coagulation

5.4. Antenatal and Intrapartum management of women identified to be at greater risk of PPH:

- At the booking appointment women should be referred for consultant led care if they have had x2 PPH (Clinical judgement should be used on an individualised basis, and if the women has had x1 significant PPH referral for consultant opinion is advised).
- Clinicians must be aware of risk factors for PPH and should take these into account when counselling women about place of delivery. Women with known risk factors for PPH should be delivered in a hospital with a blood bank on site.
- Appropriate management of antenatal anaemia may reduce the morbidity associated with PPH. Anaemia should be investigated and treated appropriately in a timely manner.

- Women should be given evidence based information about the management of the 3rd stage of labour in the antenatal period. They should be encouraged to make an informed decision about the management of 3rd stage of labour.
- Clinicians should be aware that risk factors for PPH may arise in the antenatal or intrapartum periods, care plans should be modified as and when risk factors are identified.
- Antenatal anaemia should be investigated and treated appropriately as this may reduce the morbidity associated with PPH.
- Women at significant risk of PPH should have IV access in active labour and an FBC and G+S bloods should be sent to the lab with results chased in a timely manner.
- Women at significant risk of PPH should be reviewed by the SAS Obstetric Doctor or Consultant Obstetrician when in established labour and a management plan must be documented.

5.5. **Reducing the risk of PPH at the time of delivery:**

- Prophylactic uterotonics should be routinely offered in the management of the third stage for all women as they reduce the risk of PPH
- Routine uterine massage is of **no** benefit in the prophylaxis of PPH
- For women with no risk factors for PPH delivering vaginally the uterotonic of choice is 10 units syntometrine given IM (Unless in hypertension / hypertensive disorders of pregnancy, then use 10 units Oxytocin instead).
- For women delivering by caesarean section, oxytocin (5 units by slow intravenous injection) should be used to encourage contraction of the uterus and to decrease blood loss. Carbetocin is also available.
- In women at increased risk of haemorrhage, it is possible that a combination of preventative measures may be superior to syntometrine alone to prevent PPH.
- Clinicians should consider the use of I.V. Tranexamic acid 1 gm in addition to oxytocin, at Caesarean section to reduce blood loss in women at increased risk of PPH.

5.6. **PPH Management**

Management of PPH requires multidisciplinary approach.

The anaesthetist plays a crucial role in maintaining haemodynamic stability, and if necessary, in determining the most appropriate method of anaesthesia

It should be recognised that in the event of a PPH most of the following steps will be taken simultaneously with each member of the team taking a different role.

Level of staff to be attend to woman present with minor PPH (500-1000ml) without clinical shock: The midwife in charge and the first on-call obstetric and anaesthetic staff.

Level of staff to be attend to women with major PPH (blood loss more than 1000ml) and ongoing bleeding or clinical shock: Multi-disciplinary team involving senior members of staff involving an experienced midwife, the SAS Obstetric Doctor, SAS anaesthetist, the oncall clinical haematologist with experience in major haemorrhage and porters for delivery of blood. Furthermore, the consultant obstetrician and consultant anaesthetist should be alerted and the blood transfusion laboratory should be alerted. The consultant must attend if asked.

Communication with the patient and her birthing partner is important, and clear information of what is happening should be given from the outset.
Clinicians should be aware that the visual estimation of peripartum blood loss is inaccurate and that clinical signs and symptoms should be included in the assessment of PPH

Call for Help

- Pull the emergency buzzer, state the emergency and request an emergency 2222 call and for that person to tell you when they have done so.
- Ensure someone is bringing the emergency PPH box and the emergency drugs from the drugs fridge in the treatment room on Labour Wward
- Where possible allocate a scribe and strive to use the PPH Proforma

Monitoring and Investigations:

- Vital Signs: Respiratory Rate, BP, O_2 saturations, pulse and temperature
- Obtain IV access as a matter of emergency, 2 large gauge cannulae if possible. Obtain bloods for FBC, U&Es, LFTs, coag (INR, APTT & fibrinogen) & calcium and X-match at least four units. Specify urgent X-Match. (Group specific blood can be issued by the blood bank within 15 minutes of sample receipt) (Trust guideline – Blood transfusion policy / Massive Blood Loss Policy REF)
- Insert Foley Catheter and commence strict fluid balance chart
- Weigh pads and incos – calculate estimated blood loss
- Keep Nil by Mouth

Arrest the Bleeding:

- Identify the cause of bleeding
- **Tone** – Deliver the placenta, uterine massage, insert urinary Foley's catheter to allow the uterus to involute, expel clots if necessary, consider bimanual compression
- **Trauma** – Assess the genital tract (vagina and cervix and perineum) for trauma and aim to repair promptly. If 3rd or 4th degree tear is suspected arrange for timely transfer to theatre.
- **Tissue** – Check placenta and membranes and inform the SAS Obstetric Doctor or Consultant Obstetrician if the placenta and membranes are incomplete or ragged.
- **Thrombin** – Look at the blood that has been passed, is it clotting as you would expect? If it is not consider abnormalities in clotting. In most cases abnormalities with clotting are known about prior to delivery.
- **Is surgical management indicated**? – Prepare for urgent transfer to theatre. (See surgical management heading below)

Maternal Resuscitation:

- Lay the bed flat
- **Airway** Maintenance
- **Breathing** – facial O^2 10-15L
- **Circulation** – fluid resuscitation (see below)

- Keep the patient warm – Warm towels, blankets and bair hugger.

Drugs:
- IM Syntometrine (1 millilitre)
- IM Ergometrine (500 micrograms)
- IV oxytocin (40units in 500mls normal saline at 125mls per hour over 4 hours)
- Carboprost (Hemabate) (250 micrograms deep IM injection, can be given up to a maximum of 8 times (2mg) with 15 minute intervals. After 2nd dose of Carboprost (Hemabate), transfer to theatre indicated)
- Misoprostol (1000 micrograms PR)
- Tranexamic Acid (1g IV slow bolus)

Fluid Resuscitation:
- Up to 2 litres of rapid crystalloid infusion – plasmalyte (warmed if possible)
- Blood products: if no X-matched blood available use O Rh Neg or group specific blood
- Cell salvage should be used if at Caesarean Section and staff skills present
- Blood products (Fresh Frozen Plasma (FFP), Cryoprecipitate and Platelets – discuss with consultant haematologist)
- The best equipment available should be used to achieve rapid warmed infusion of fluids.
- Special blood filters should not be used, as they slow infusions.

Target Blood Values:
- Haemoglobin >80 g/L
- Platelets >75 x 10^9/L
- INR or APTTR <1.5
- Fibrinogen >2g/L or >200mg/dL

If bleeding exceeds 1500mls and is on-going commence the major obstetric haemorrhage protocol (Appendix B)
NOTE: If the woman is pre-eclamptic or hypertensive, always take this into consideration in the management of the third stage and/or the management of a PPH. If the woman is on a fluid restriction see guidance for the use of 40 units oxytocin in 50mls saline through a syringe driver (Hypertension in pregnancy Trust guideline)

Surgical Management:
- Uterine Tamponade Balloon: This is the first line surgical intervention for most women where uterine atony is the only or main cause of haemorrhage (success in 91% of women).
    - see Appendix A for technique
- Bilateral uterine artery ligation (O' Leary suture) and to further diminish blood flow to the uterus, sutures can also be placed across the vessels within the utero-ovarian ligaments (success in 92% of women when used as a second-line approach). (ACOG Practice-bulletin 183)

- Uterine Compression suture: B-Lynch Suture (success in 76% of women)– see Appendix A for technique
- Embolisation: Only available as an elective procedure at NDDH by interventional radiology (success in 86.5% of women)
- Hysterectomy - this decision may only be taken by the on-call Consultant after discussion with a second Consultant

On-going Care:

- A clear plan of care should be written by the SAS Obstetric Doctor or Consultant Obstetrician and the location of the care required should be considered e.g. ITU
- Women should receive 1:1 care in a high dependency environment
- VTE assessment – plan regarding time of LMWH administration in view of excessive blood loss
- Strict fluid balance chart
- Hourly urine output with urometer if EBL >1000mls and catheter care
- Risk assessment booklet updated and management plan updated if risk of pressure ulcer is raised, aim to mobilise when appropriate
- Closely observe lochia and uterine tone
- Have a conversation with mother / partner around potential impact of PPH on lactation and ensure appropriate feeding support in place
- Clear plan for timing for any repeat bloods – usually following morning in a minor PPH, after 6hrs if major PPH.
- Analgesia
- MEOWS at least hourly until stable, then 4 hourly for 24 hrs.
- Obstetric review by the SAS Obstetric Doctor or Consultant Obstetrician prior to transfer to postnatal ward
- **Document –** all procedures should be documented contemporaneously. Where possible the PPH proforma (Appendix C) should be utilised. If it is not possible to use this, documentation should include: personnel present, procedures performed, drugs administered and observations taken.
- **Datix –** An incident form should be completed in the event of a minor, major or massive PPH.
- **Debrief –** Debrief is a useful tool for both staff and the women and families in our care. Efforts should be made to ensure the woman and her family are reassured throughout the event and that the opportunity to formally debrief should be given.

6. Monitoring Compliance with and the Effectiveness of the Guideline

6.1. An up to date copy of this guideline is available to all staff on the Trust intranet. The Flow charts in Appendix B should be laminated and posted in plain view in all clinical areas. As a matter of routine, this guideline will be reviewed triennially by the Maternity Services Guideline group.

6.2. Reporting for non-compliance and review of effectiveness of the guideline will be identified through the risk process within maternity and led by appointed maternity Risk leads. The maternity services audit process will include review of this guideline. All versions of these guidelines will be archived in electronic format by the author within the Maternity Team policy archive. Any revisions to the final document will be recorded on the Document Control Report. To obtain a copy of the archived guidelines, contact should be made with the Maternity team.

7. References

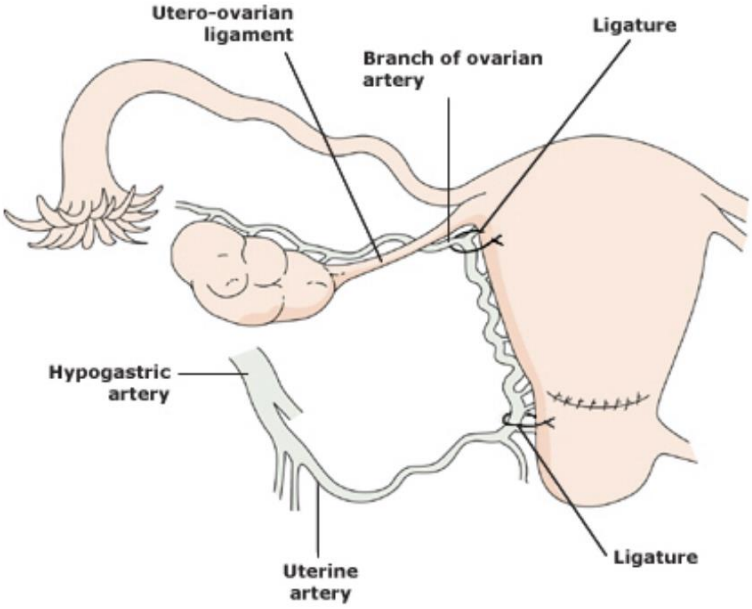
- Saving lives, Improving Mothers' care, Maternal, Newborn and Infant Clinical Outcome Review Programme, MBRRACE UK, December 2016
- Thompson AJ, Ramsay JE, Rich D. Antepartum Haemorrhage. Royal College of Obstretcians and Gynaecologist's Green top guideline No.63. 2011
- Mayrides E, Allard S, Chandraharam E, Collins P, Green L, Hunt BJ, Riris S, Thomson AJ. Prevention and management of postpartum haemorrhage. Royal College of Obstetricians and Gynaecologist's Green top guideline No 52. 2016
- Nursing and Midwifery Council: The Code
- World Health Organisation.
- National Institute for Clinical Excellence. Intrapartum Care for Healthy Women and Babies. 2017
- Trust guideline- Blood transfusion
- Trust guideline – Massive Blood Loss Transfusion Policy
- Trust guideline- Normal Labour care
- Trust guideline- Hypertension in pregnancy
- ACOG practice bulletin No. 183, 2017: PPH
- UpToDate; 2021 : PPH

8. Associated Documentation

- PPH Proforma
- Major obstetric haemorrhage protocol
- Fluid Resuscitation in major obstetric haemorrhage

9. Appendix A: Procedures for PPH:

Procedure for Uterine Artery Ligation O'Leary Suture (image from UpTodate)

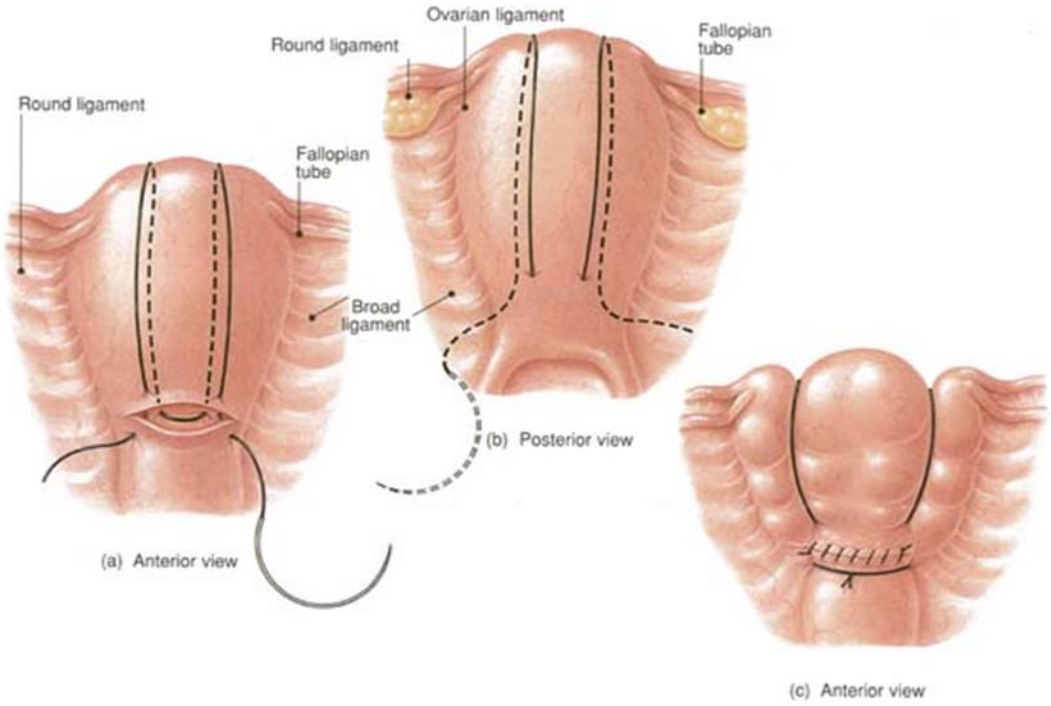


Sutures are placed to ligate the ascending uterine artery and the anastomotic branch of the ovarian artery. The procedure should be performed on both sides.

Procedure for insertion of B-Lynch Suture
Obstetric competency required: NDDH training annually via PROMPT day.

1. Laparotomy required if Caesarean has not taken place.
2. Exteriorise uterus
3. Bi-manual compression to assess the likelihood of success of the B-Lynch suture.
4. 70mm round bodied hand needle with a No.2 vicryl suture
5. Puncture uterus 3cm below the right lower edge of the uterine incision and 3 cm medial to the right lateral border.
6. Bring needle back out 3cm above the upper incision margin and about 4cm medial to the lateral border
7. Pass over the fundus about 3 – 4 cm from the right cornual border.
8. Pull suture under moderate tension assisted by manual compression by the assistant.
9. Pass the needle through the posterior wall of the uterus at the same level as the upper anterior entry point and place a suture across the posterior wall exiting at the same position on the left side
10. Pass the suture over the fundus about 3-4cm from the left cornual border and enter the uterus above and below the incision as for the first side.
11. Pull the two ends of the suture
12. Tie with a double throw on the first knot followed by two further knots.
13. (Close lower transverse uterine incision)

Parts (a) and (b) demonstrate the anterior and posterior views of the uterus showing the application of the B-Lynch Brace suture. Part (c) shows the anatomical appearance after competent application. (B-Lynch et al, 1997)



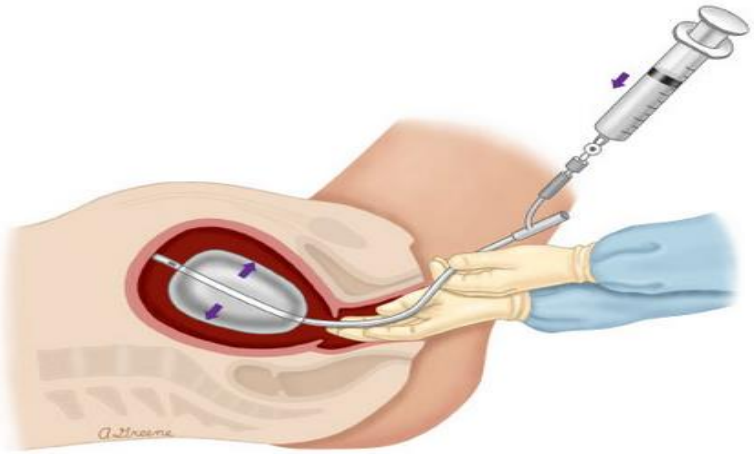
Uterine Balloon Tamponade technique

After vaginal delivery

1. Before inserting the balloon, ensure that the bladder is empty by inserting a Foley catheter.
2. Inspection of the uterine cavity should be made to ascertain that the uterine cavity is clear of placental fragments.
3. The cervix and vagina should be cleansed with an antiseptic solution, such as povidone iodine.
4. Grasp the cervix with ring forceps. Insert the balloon into the cavity of the uterus, making sure that the entire portion of the balloon passes the cervical canal above the internal cervical os.
5. Once the correct placement is confirmed, inflate the balloon with sterile saline using the enclosed syringe. The recommended maximum capacity of the balloon is 500 mL.
6. In order to maximize the effect of tamponade most notable to the lower uterine segment, apply gentle traction to the shaft of the balloon. This can be achieved and maintained by securing the balloon shaft to the patient leg or attaching to a weight.
7. Monitor patient continuously for signs of increased bleeding and uterine cramping.

After caesarean section

1. Insert the balloon end of the catheter through the open uterine incision to the cervix and then place the port end into the vagina.
2. Close the uterine incision while taking special care not to damage the balloon by the suturing needle.
3. Inflate the balloon under direct visualization.
4. An alternative approach is to close the uterus first and then insert the balloon from the vagina and inflate it while the surgeon watches from above.



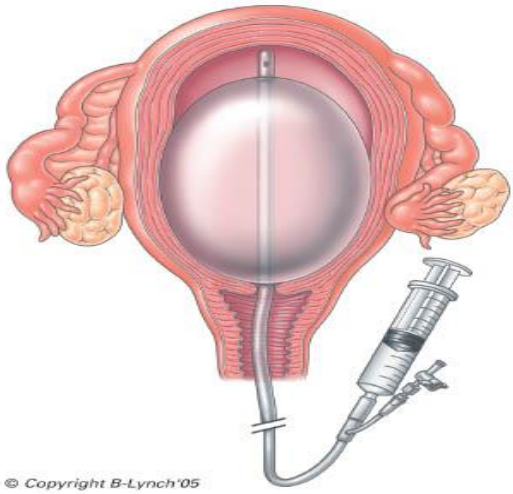


Figure 2 Rüsch hydrostatic balloon catheter

10. Appendix B: Fluid Resuscitation and Flowcharts

Massive Obstetric Haemorrhage (Antepartum >24 weeks gestation) – Page 1

Definition: Blood loss > 1000 mls (or class III / IV shock)

Call for help
- Dial 2222. Say: "Emergency obstetric team" and state location
- Inform obstetric and anaesthetic duty consultants

ABC assessment

Left lateral position & give oxygen (non-rebreathe mask 15 L/min)

- IV access: 2 large-bore cannulae
- Bloods: FBC, U&Es, LFTs, coag (INR, APTT & fibrinogen) & calcium; **Crossmatch 4 units RBCs**
- IV Fluid: Warmed (up to 2 litres Plasmalyte then consider blood products)
- Auto-transfuse: Head down tilt
- Observations: Pulse, BP, respiratory rate, O2 sats & temperature
- Assess for other signs of shock e.g. agitation / confusion / loss of consciousness, cool pale moist skin
  - **Clinical shock is an important indicator of maternal blood loss. Bleeding may be concealed.**
- Assess abdomen for pain, tenderness & rigidity
- Weigh all swabs & estimate blood lost

Assess fetus: Fetal compromise is an indicator of blood loss

Is immediate delivery of fetus necessary? (maternal or fetal compromise)

Blood products
- Bleep 045 to alert blood transfusion laboratory. State "Major Obstetric Haemorrhage"
- Initially 4 units red cells (crossmatched, group specific or "O Rh D negative" as appropriate)
- For further details **See blood component box overleaf**
- Tranexamic acid 1 gram IV slow bolus

Decide mode of delivery
- Consider cause: Placenta previa, placental abruption, uterine rupture, vasa previa
- Usually C-section unless in established labour

Continue overleaf

Other Monitoring & Investigations
- Keep woman warm:
  Warmed fluid & bair hugger
- HemoCue and blood gas
- Urinary catheter & urine output
- Consider central and arterial lines
- Commence obstetric MEOWS chart

Massive Obstetric Haemorrhage (Antepartum >24 weeks gestation) – Page 2

Continuation from overleaf

At delivery

- Beware of coagulopathy associated with placental abruption
- Are pH, temperature & clotting normal?
- Beware of significant risk of PPH
- If C-section, follow C-section flowchart

Postpartum

- Consider HDU / ICU
- Antibiotics if uterus or vagina explored
- **Thromboprophylaxis assessment** when bleeding controlled

Blood component therapy

Red cells, FFP & platelets:

- Initially request **4 units red cells** for major obstetric haemorrhage
- If required, then request **major blood loss pack 1** (**4 units red cells + 4 units FFP**)
- **NDDH does NOT stock platelets.** Requesting major blood loss pack 1 will trigger 2 units of platelets to be ordered. Delivery time may exceed 2 hours.
- Consider seeking advice from consultant haematologist
- If required then give **major blood loss pack 2** (**4 units red cells, 4 units FFP, 2 units cryoprecipitate & platelets** if available)
- See NDDH massive blood loss policy for further details

Targets:

- Aim Hb > 80 g/L; platelets >75 x 10^9/L; INR < 1.5; fibrinogen > 200mg/dl

American College of Surgeons Classes of Acute Haemorrhage				
Factors	**1**	**2**	**3**	**4**
Blood Loss (%)	<15	15-30	30-40	>40
Blood Loss (ml)	<750	750-1500	1500-2000	>2000
Pulse	>100	>100	>120	>140
Blood Pressure	Normal	Normal	↓	↓↓
Pulse Pressure	N or ↓	↓	↓↓	↓↓
Capillary refill (s)	<2	2-3	3-4	>5
Respiratory Rate	14-20	20-30	30-40	>40
Urine Output (ml/hr)	≥30	20-30	5-10	Negligible
Mental status	Slightly anxious	Mildly anxious	Anxious / confused	Confused / lethargic

Please remember the impact of different levels of haemorrhage on different maternal booking weights. Smaller patients will be more greatly impacted by a seemingly smaller haemorrhage.

Massive Obstetric Haemorrhage (Postpartum) – Page 1

Definition: Blood loss > 1500 mls (or class III / IV shock)

Call for help

- Dial 2222. Say: "Emergency obstetric team" and state location
- Inform obstetric and anaesthetic duty consultants

ABC assessment

Lie flat & give oxygen (non-rebreathe mask 15 L/min)

- Rub up contraction & expel clots +/- Bimanual compression
- Deliver placenta & membranes. Apply pressure to tears.

- IV access: 2 large-bore cannulae
- Bloods: FBC, U&Es, LFTs, coag (INR, APTT & fibrinogen) & calcium; **Crossmatch 4 units RBCs**
- IV Fluid: Warmed (up to 2 litres Plasmalyte then consider blood products)
- Auto-transfuse: Elevate legs +/- head down tilt
- Observations: Pulse, BP, respiratory rate, O2 sats & temperature

- Assess cause: **4 T**s (Tone; Tissue; Trauma; Thrombin)
- Weigh all swabs & estimate blood lost

Stop the bleeding

TONE

- Empty bladder
- Uterotonic drugs:
  - Syntometrine 1ml IM – NOT if high BP

  OR
  - Oxytocin 5 units IV or 10 units IM
  - Ergometrine 500 micrograms IM or slow IV – NOT if high BP

  THEN
- Oxytocin infusion – 40 units in 500 ml 0.9% saline; 125 ml/h
- Carboprost (Hemabate) 250 micrograms IM every 15 mins up to 8 times – NOT if asthmatic
- Misoprostol 1000 micrograms rectally

TISSUE

Check placenta & membranes

TRAUMA

Suture any tears

THROMBIN

Check for signs of clotting disorder

Blood products

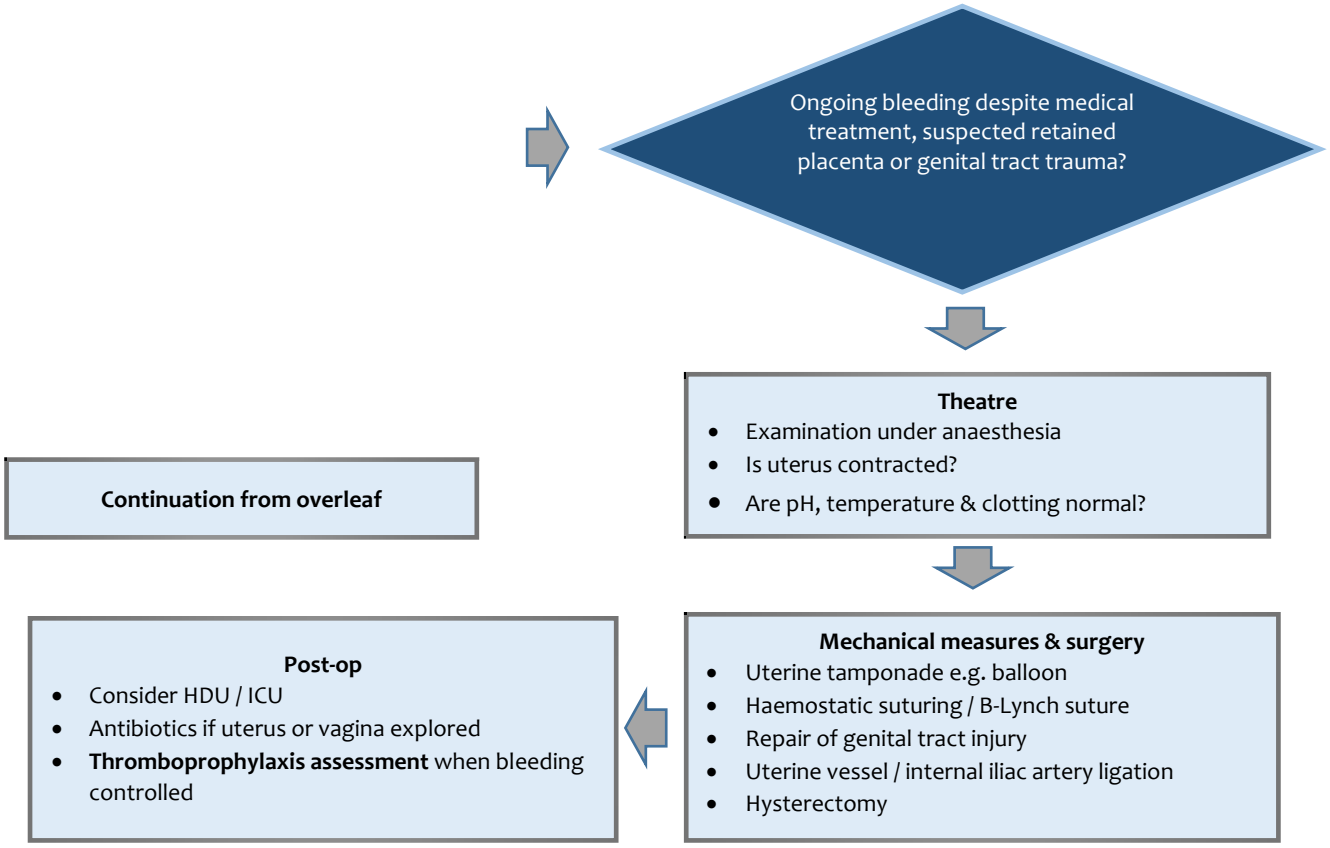
- Bleep 045 to alert blood transfusion laboratory
- Initially 4 units red cells (crossmatched, group specific or "O Rh D negative" as appropriate)
- For further details **See blood component box overleaf**
- Tranexamic acid 1 gram IV slow bolus

Other Monitoring & Investigations

- Keep woman warm: Warmed fluid & bair hugger
- HemoCue and blood gas
- Urinary catheter & urine output
- Consider central and arterial lines
- Commence obstetric MEOWS chart

Continue overleaf

Massive Obstetric Haemorrhage (Postpartum) – Page 2



Blood component therapy

Red cells, FFP & platelets:

- Initially request **4 units red cells** for major obstetric haemorrhage
- If required, then request **major blood loss pack 1** (**4 units red cells + 4 units FFP**)
- NDDH does NOT stock platelets. Requesting major blood loss pack 1 will trigger 2 units of platelets to be ordered. Delivery time may exceed 2 hours.
- Consider seeking advice from consultant haematologist
- If required then give **major blood loss pack 2** (**4 units red cells, 4 units FFP, 2 units cryoprecipitate & platelets** if available)
- See NDDH massive blood loss policy for further details

Targets:

- Aim Hb > 80 g/L; platelets >75 x 10^9/L; INR < 1.5; fibrinogen > 200mg/dl

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Capillary refill (s)	<2	2-3	3-4	>5
Respiratory Rate	14-20	20-30	30-40	>40
Urine Output (ml/hr)	≥30	20-30	5-10	Negligible
Mental status	Slightly anxious	Mildly anxious	Anxious / confused	Confused / lethargic

Please remember the impact of different levels of haemorrhage on different maternal booking weights. Smaller patients will be more greatly impacted by a seemingly smaller haemorrhage.

11. Appendix C: PPH Proforma (see overleaf)

NHS
Northern Devon Healthcare
NHS Trust

Mother's Name: _____

Hospital Number: _____

DOB: _____

Or attach addressograph label

OBSTETRIC HAEMORRHAGE PROFORMA

To be used for PPHs equal to or more than 1500 mls or clinical concern

Date:	Time:	Name of person completing form:

Call for help (Labour Ward/2222) **Time called:**

ACTIVATE MAJOR OBSTETRIC HAEMORRHAGE PROTOCOL	Time Activated:	:	NOT CLINICALLY REQUIRED: ☐
1500 - 2000mls and ongoing haemorrhage			

Key Staff	Name	Time arrived/ informed
Labour Ward Coordinator (Bleep 025)		
Obstetric Consultant (Bleep 800 / Phone)		
Obstetric SAS Doctor (Bleep 013)		
Primary Midwife		
Anaesthetic Consultant (Phone)		
Labour Ward Anaesthetist (Bleep 822)		
Blood Bank (Bleep 045)		
On Call Haematologist		
Porter		

RESUSCITATION /INITIAL MANAGEMENT

ACTION	TIME	ACTION	TIME
High flow $O_2 \geq 10$L/min		Examination (Abdominal/ Vaginal/ Placenta)	
IV Access large bore cannula 1		Massage Uterus	
large bore cannula 2		Bimanual compression	
Observations (MEOWS)		Catheter- Foley's with hourly urine measurement	
Rapid IV fluid resuscitation – detail overleaf		Patient warmer/ Bair hugger	
Start Critical Care chart		Other (Specify):	

BLOOD SAMPLES	TIME	RESULT	BLOOD SAMPLES	TIME	RESULT
FBC / Haemocue			U&Es		
Clotting			Xmatch 4 units (if no MOH Protocol)		
Group & Save (if not already taken)			Other (Specify):		

IDENTIFY CAUSE(S) OF BLEEDING – tick cause(s)

Tone ☐	Tissue ☐	Trauma ☐	CloTTing ☐

DRUGS	Dose	Time	Order	Dose	Time
Syntometrine 1st dose (5 IU/ 500mcg IM)					
Syntometrine 2nd dose / Ergometrine					
Oxytocin Bolus (5 IU IV or 10 IU IM)					
Oxytocin Infusion (40 IU in 500 ml 0.9% saline IV)					
Haemobate (250mcg IM)	1			5	
	2			6	
	3			7	
	4			8	
Tranexamic acid (1g IV)					
Misoprostol (600 – 1000 mcg PR)					
Calcium gluconate 10% (10ml IV - Anaesthetist)					

CRYSTALLOID FLUIDS GIVEN

Type	Amount Infused	Warmed?	Time Started	Time completed
		Y / N		
		Y / N		
		Y / N		

Cell Saver – Used / Not Used. If used volume replaced:

BLOOD AND BLOOD PRODUCTS GIVEN

O -ve, group specific, X-matched, FFP, cryoprecipitate, platelets, fibrinogen concentrate, other

BLOOD LOSS (running total)

Type	Volume/Unit(s)	Time Started	Time EBL measured	Swabs Weighed	EBL
				Total	Total

FURTHER MANAGEMENT

	Performed	Order Performed	Date	Time
EUA +/- genital tract repair				
Manual removal of placenta				
Arterial/ CVP line				
Intra-Uterine balloon tamponade				
B Lynch suture				
Bilateral ligation Of Uterine Arteries				
Hysterectomy				
Interventional Radiology				

POST-NATAL CARE

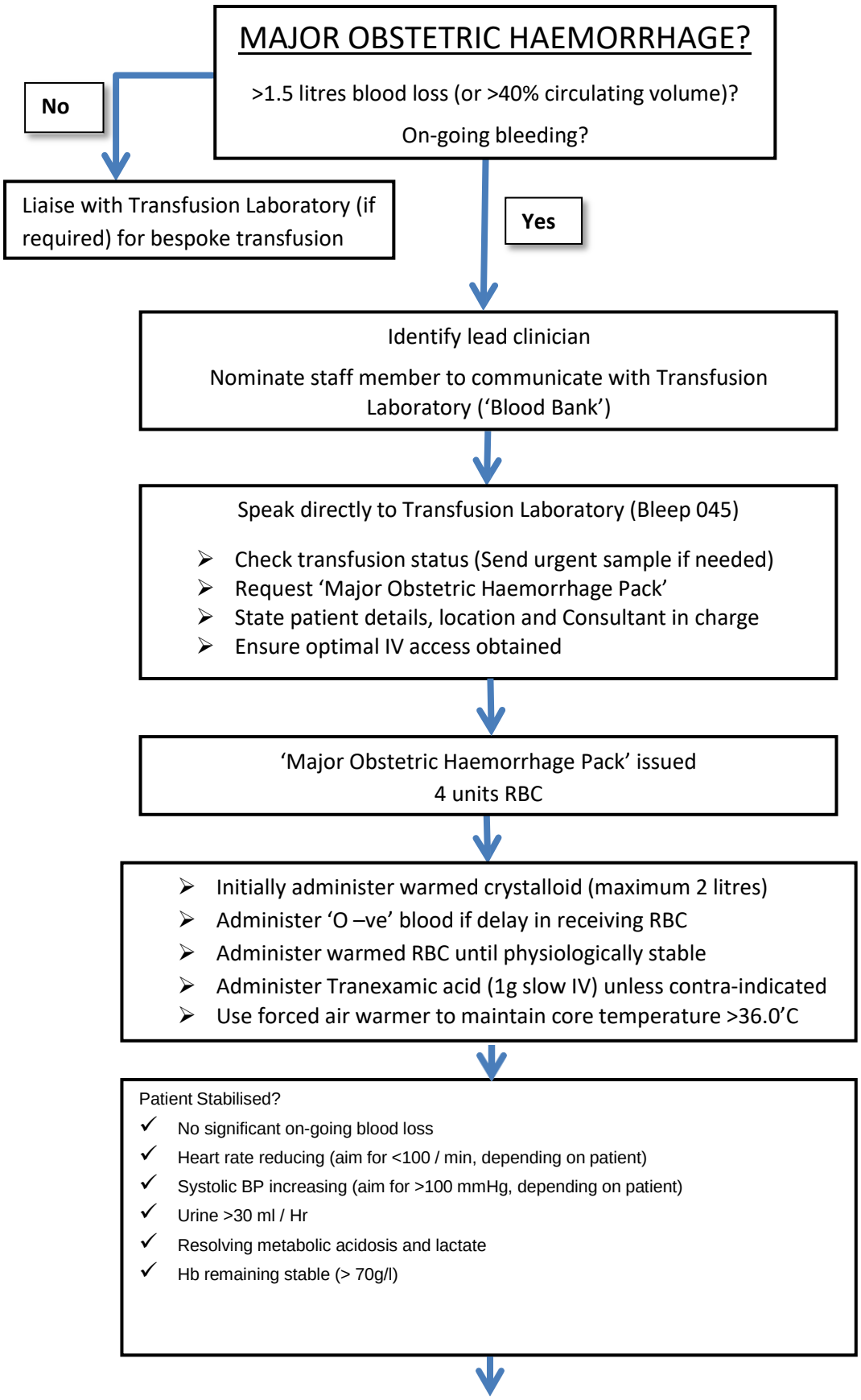
Level of care:	Labour Ward ☐	HDU ☐	ICU ☐
Blood Tests:	FBC ☐	U&E ☐	Coag ☐
Thromboprophylaxis:	TEDS ☐	LMWH ☐	Flowtrons ☐
Urometer:			
Antibiotics (Dose/ Time):			
MEOWS/ HDU chart:			
Fluids/ Blood:			
Duty of Candour:			
DATIX Reference:			
Any Other Information:			

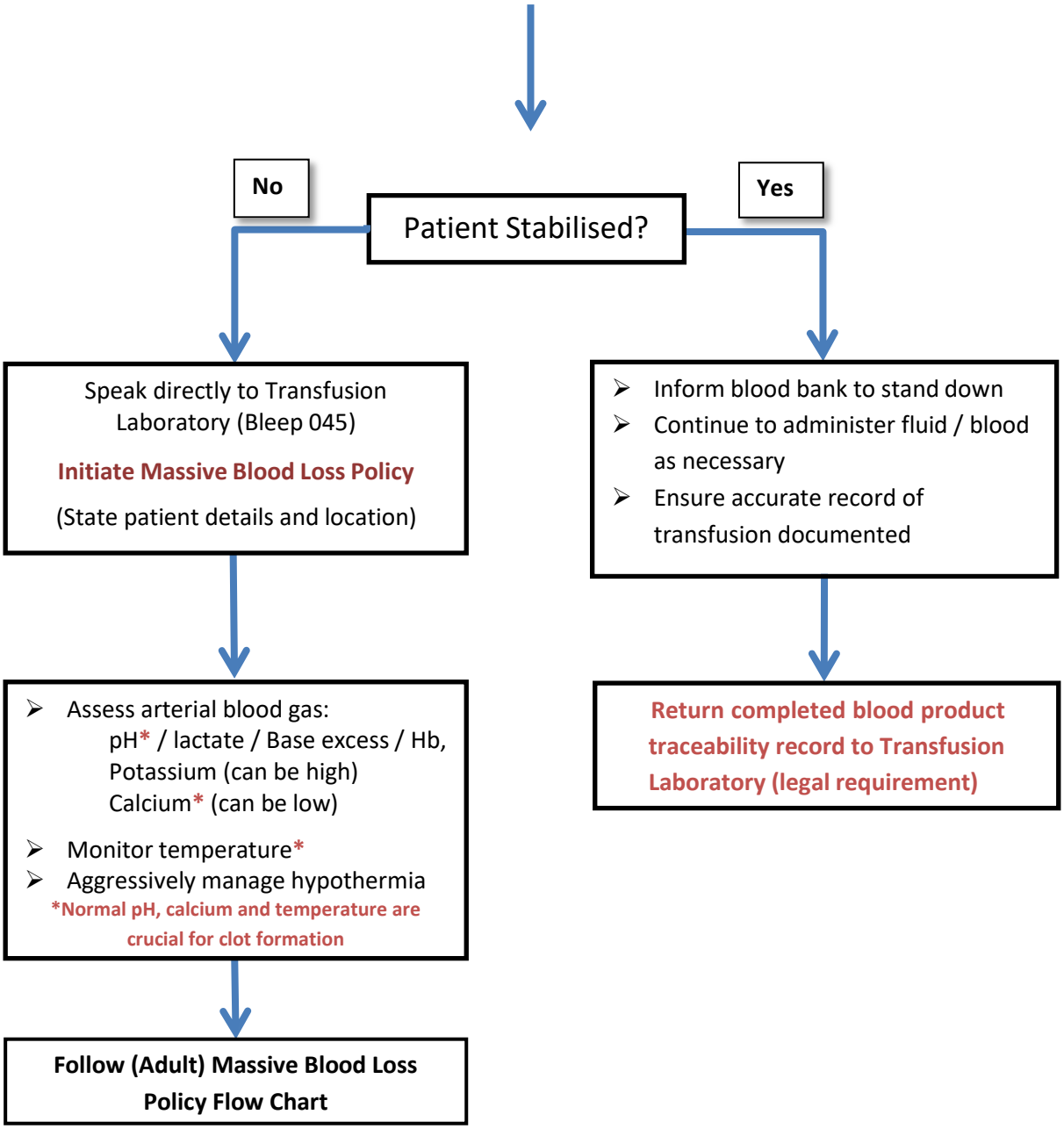
DEBRIEF

What went well?

What could have gone better?

12. Appendix D: Major Obstetric Haemorrhage





Massive Blood Loss Policy from the Hospital Transfusion Team

("NDHT-MBL-policy-v4.0-Oct-2020-flowcharts-included.pdf")